

Initiating Search

November 6, 2025, 11:07 AM

• Search:

CAS Lexicon Search:

Anhydrides - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (17,127)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,767)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	

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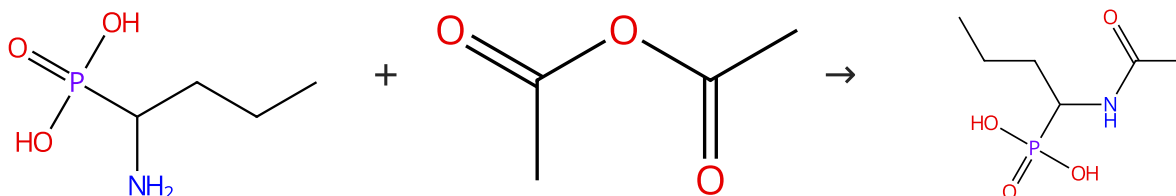
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (21)

Suppliers (84)

Supplier (1)

31-366-CAS-2729156

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

By: Kudzin, Z. H.; et al

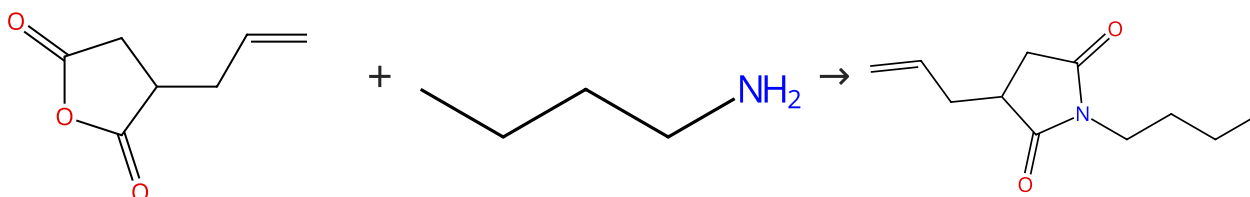
Polish Journal of Chemistry (2005), 79(3), 499-513.

1.1 Reagents: Acetic acid

Solvents: Acetic acid; 30 min, 100 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (61)

Suppliers (82)

Suppliers (2)

31-366-CAS-2943251

Steps: 1 Yield: 100%

New succinamic acids, -γ-lactones, and -succinimides from (3-perfluoroalkyl-2-iodoalkyl)succinic acid anhydrides and amines

By: Brace, Neal O.; et al

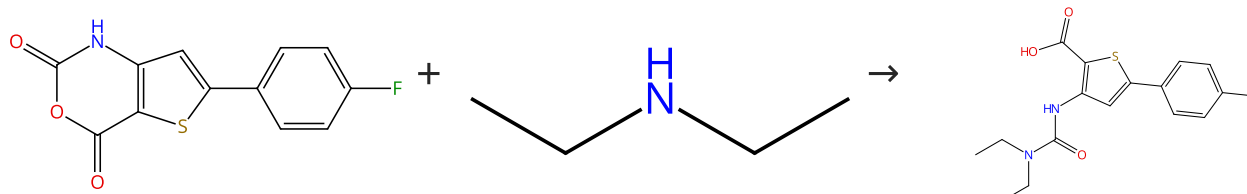
Journal of Fluorine Chemistry (2006), 127(1), 108-125.

1.1 Solvents: Dichloromethane; 15 min, 6 - 20 °C; reflux

1.2 Solvents: Toluene; 2.5 h, 116 °C; 18 h, 116 - 144 °C

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

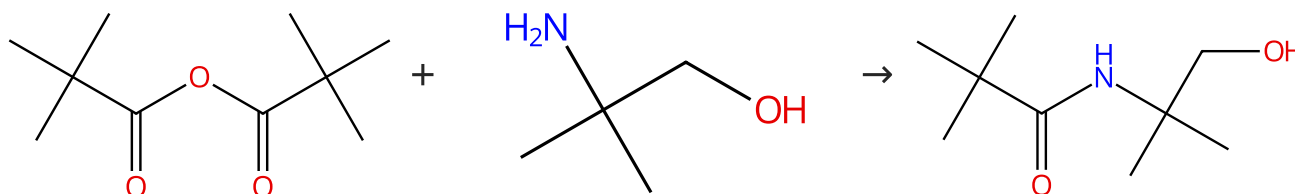
Suppliers (77)

Supplier (1)

31-366-CAS-12576406	Steps: 1 Yield: 100%	Solution-phase parallel synthesis of a 1140-member ureidothiophene carboxylic acid library
1.1 Solvents: Water; 2 h		By: Le Foulon, Francois-Xavier; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Journal of Combinatorial Chemistry (2005), 7(2), 253-257.
Experimental Protocols		

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

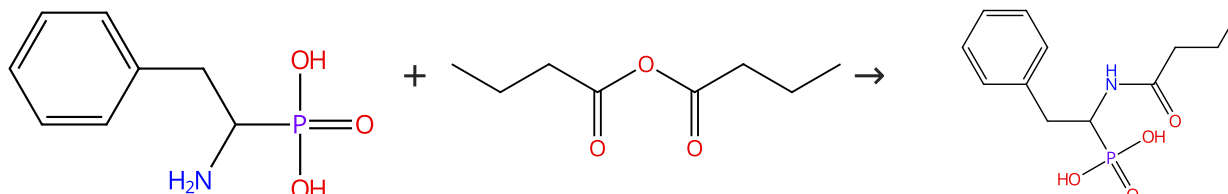
Suppliers (93)

Suppliers (4)

31-366-CAS-6122507	Steps: 1 Yield: 100%	Nucleophilic acyl substitutions of anhydrides with protic nucleophiles catalyzed by amphoteric, oxomolybdenum species
1.1 Catalysts: Molybdenum chloride oxide (MoCl ₂ O ₂), (7-4)- Solvents: Dichloromethane; rt; 30 min, rt		By: Chen, Chien-Tien; et al
1.2 1 h, rt		Journal of Organic Chemistry (2005), 70(4), 1188-1197.
1.3 Reagents: Water; cooled		
Experimental Protocols		

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (20)

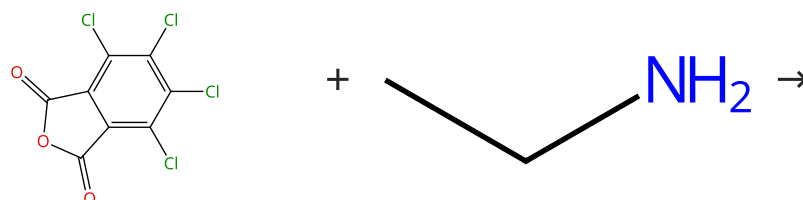
Suppliers (61)

Suppliers (2)

31-366-CAS-11441779	Steps: 1 Yield: 100%	1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids
1.1 Solvents: Water; 30 min, 100 °C		By: Kudzin, Z. H.; et al
		Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (43)

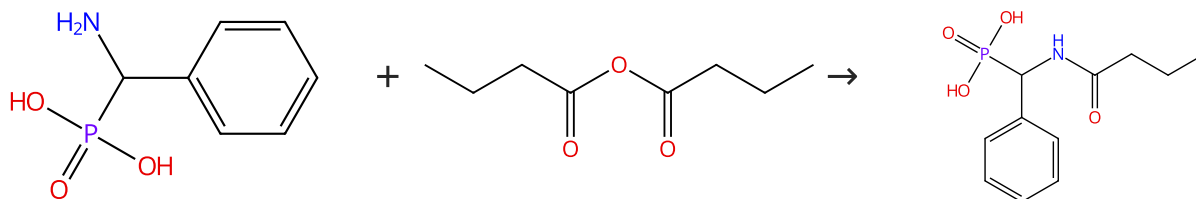
Suppliers (86)

Multi-component
structure image
available in CAS
SciFinder

31-366-CAS-3624785	Steps: 1 Yield: 100%	Quantitative Solid-State Reactions of Amines with Carbonyl Compounds and Isothiocyanates
No Data Available		By: Kaupp, Gerd; et al Tetrahedron (2000), 56(36), 6899-6911.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (23)

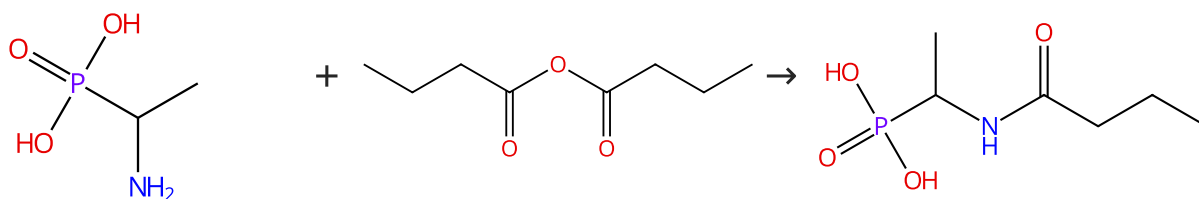
Suppliers (61)

Suppliers (2)

31-366-CAS-9342087	Steps: 1 Yield: 100%	1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids
1.1 Solvents: Water; 30 min, 100 °C		By: Kudzin, Z. H.; et al Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (49)

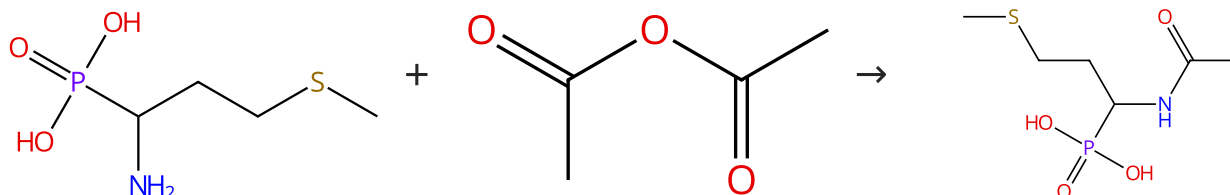
Suppliers (61)

Suppliers (3)

31-366-CAS-7110672	Steps: 1 Yield: 100%	1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids
1.1 Solvents: Water; 30 min, 100 °C		By: Kudzin, Z. H.; et al Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (5)

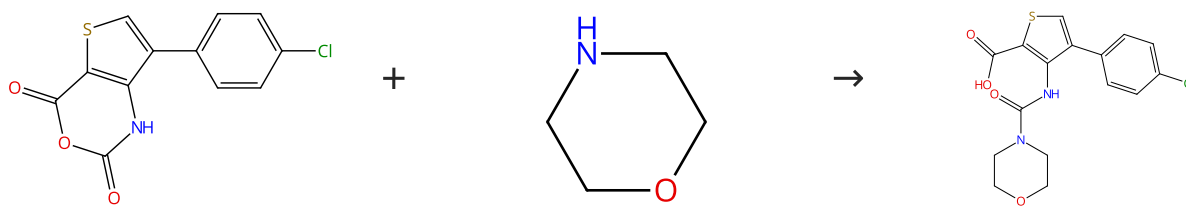
Suppliers (84)

Suppliers (2)

31-366-CAS-11427442	Steps: 1 Yield: 100%	1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids
1.1 Reagents: Acetic acid Solvents: Acetic acid; 30 min, 100 °C		By: Kudzin, Z. H.; et al Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (88)

31-366-CAS-4797698

Steps: 1 Yield: 100%

Solution-phase parallel synthesis of a 1140-member ureidothiophene carboxylic acid library

By: Le Foulon, Francois-Xavier; et al

Journal of Combinatorial Chemistry (2005), 7(2), 253-257.

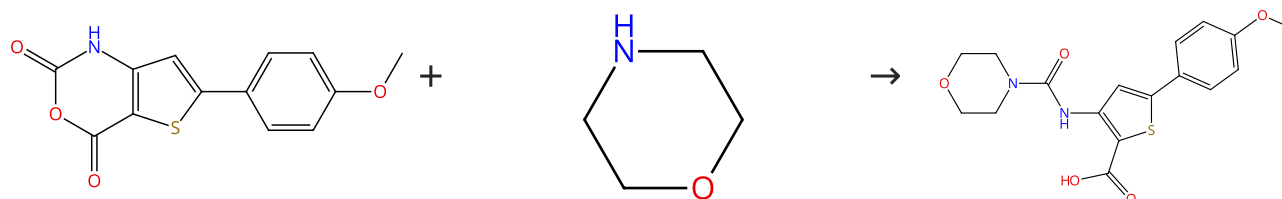
1.1 Solvents: Water; 2 h

1.2 Reagents: Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (88)

31-366-CAS-6446154

Steps: 1 Yield: 100%

Solution-phase parallel synthesis of a 1140-member ureidothiophene carboxylic acid library

By: Le Foulon, Francois-Xavier; et al

Journal of Combinatorial Chemistry (2005), 7(2), 253-257.

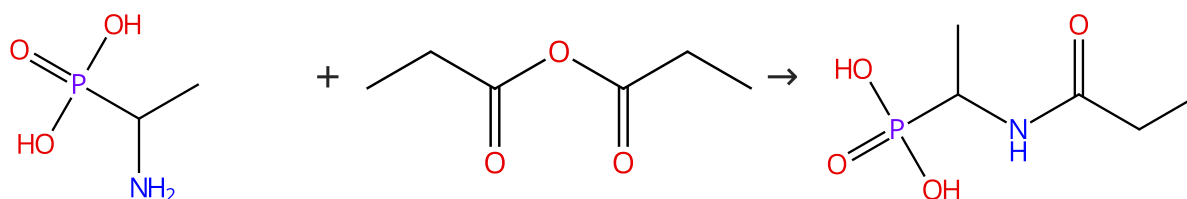
1.1 Solvents: Water; 2 h

1.2 Reagents: Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (49)

Suppliers (47)

Suppliers (7)

31-366-CAS-15506372

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

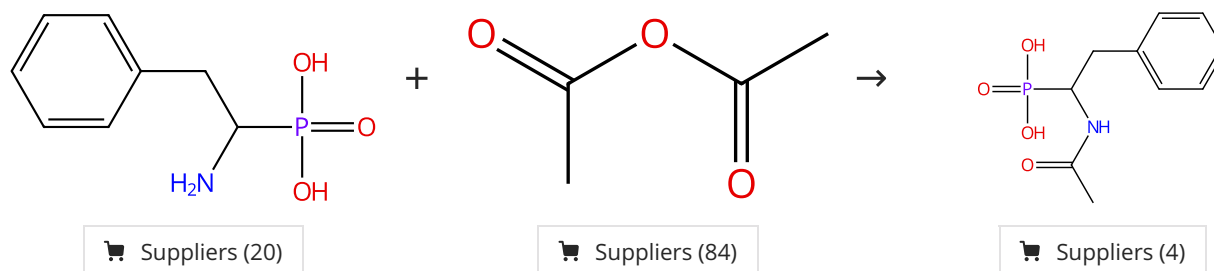
By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

1.1 Solvents: Water; 30 min, 100 °C

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



31-366-CAS-9112212

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

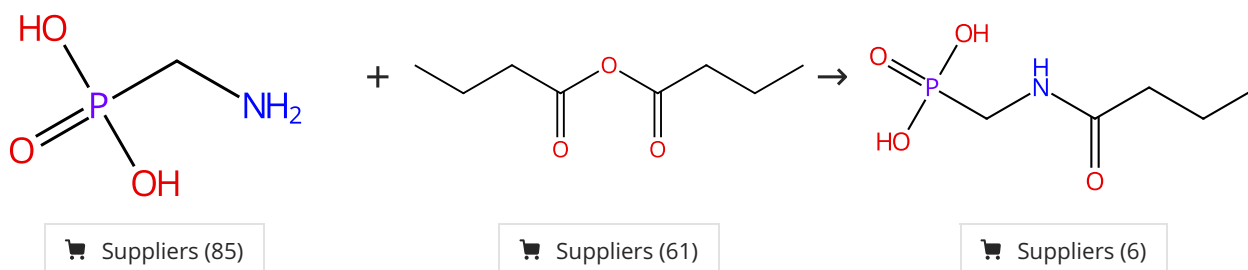
1.1 Reagents: Acetic acid
Solvents: Acetic acid; 30 min, 100 °C

By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



31-366-CAS-5049723

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

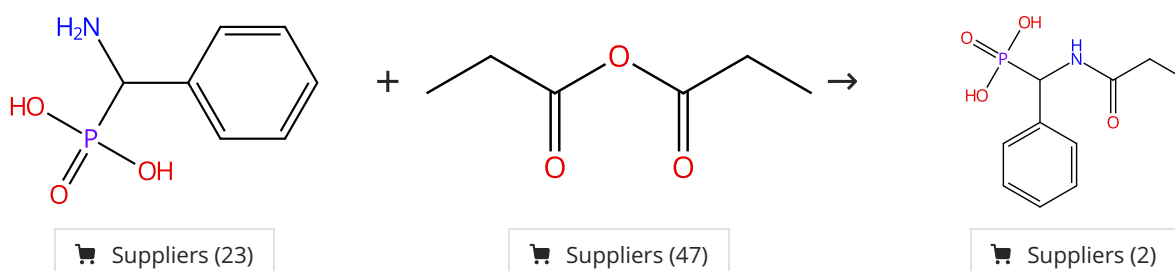
1.1 Solvents: Water; 30 min, 100 °C

By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



31-366-CAS-14215050

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

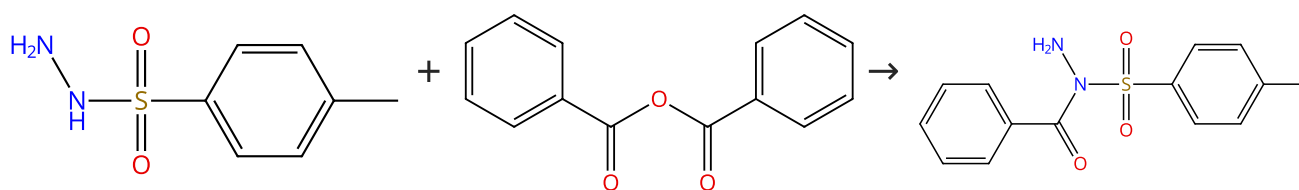
1.1 Solvents: Water; 30 min, 100 °C

By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (78)

Supplier (1)

31-366-CAS-9624167

Steps: 1 Yield: 100%

4-Aminopyridine Catalyzed Direct and Regioselective Acylation of N-Tosylhydrazide

By: Namba, Kosuke; et al

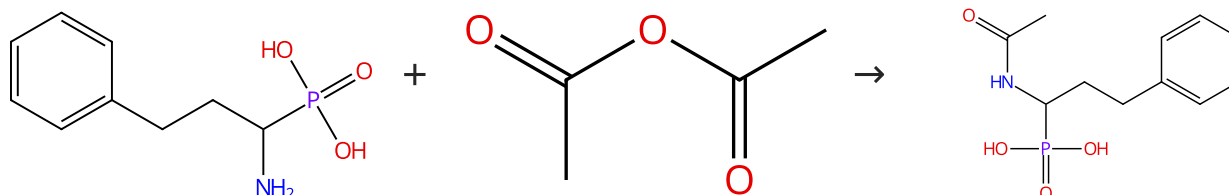
Organic Letters (2009), 11(21), 4970-4973.

- 1.1 **Reagents:** Triethylamine
Catalysts: 4-Aminopyridine
Solvents: Dichloromethane; rt → 0 °C
- 1.2 0 °C; 30 min, 0 °C
- 1.3 **Reagents:** Sodium bicarbonate
Solvents: Water

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (11)

Suppliers (84)

Suppliers (2)

31-366-CAS-11238102

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

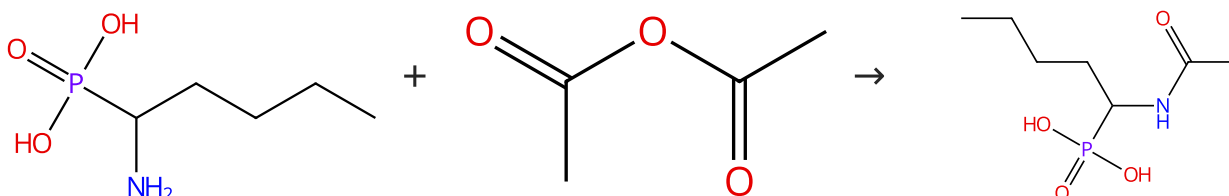
By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

- 1.1 **Reagents:** Acetic acid
Solvents: Acetic acid; 30 min, 100 °C

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (15)

Suppliers (84)

Suppliers (2)

31-366-CAS-574690

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

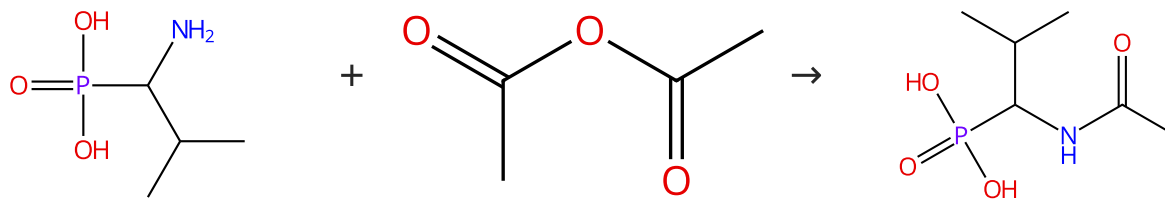
By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

- 1.1 **Reagents:** Acetic acid
Solvents: Acetic acid; 30 min, 100 °C

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (24)

Suppliers (84)

Suppliers (2)

31-366-CAS-13978354

Steps: 1 Yield: 100%

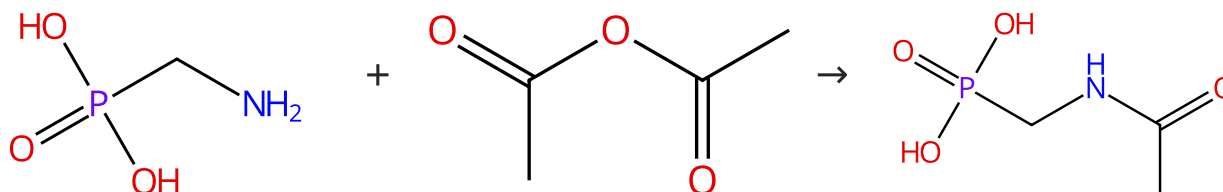
1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

1.1 Reagents: Acetic acid
Solvents: Acetic acid; 30 min, 100 °C

By: Kudzin, Z. H.; et al
Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (85)

Suppliers (84)

Suppliers (29)

31-366-CAS-309854

Steps: 1 Yield: 100%

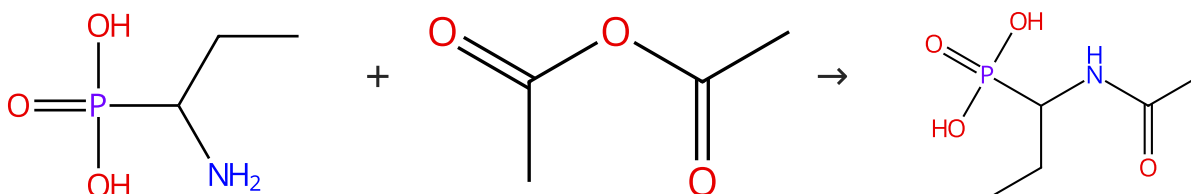
1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

1.1 Reagents: Acetic acid
Solvents: Acetic acid; 30 min, 100 °C

By: Kudzin, Z. H.; et al
Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (26)

Suppliers (84)

Supplier (1)

31-366-CAS-9292393

Steps: 1 Yield: 100%

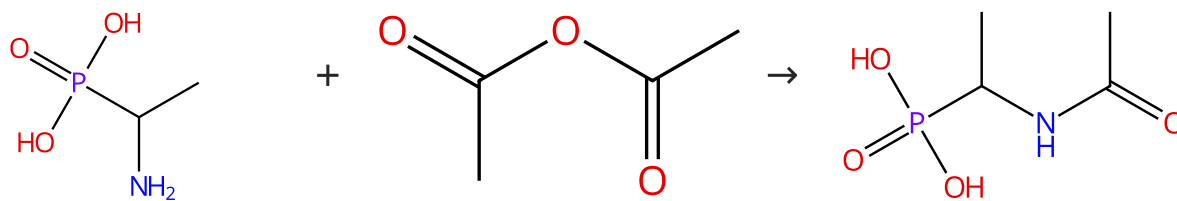
1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

1.1 Reagents: Acetic acid
Solvents: Acetic acid; 30 min, 100 °C

By: Kudzin, Z. H.; et al
Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (49)

Suppliers (84)

Suppliers (3)

31-366-CAS-2451866

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

1.1 Reagents: Acetic acid

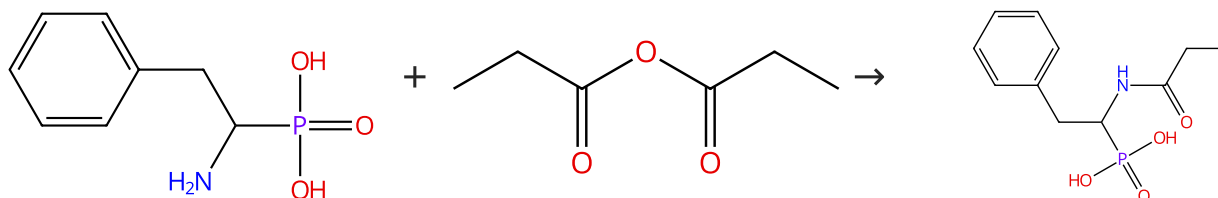
Solvents: Acetic acid; 30 min, 100 °C

By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (20)

Suppliers (47)

Suppliers (2)

31-366-CAS-2941626

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

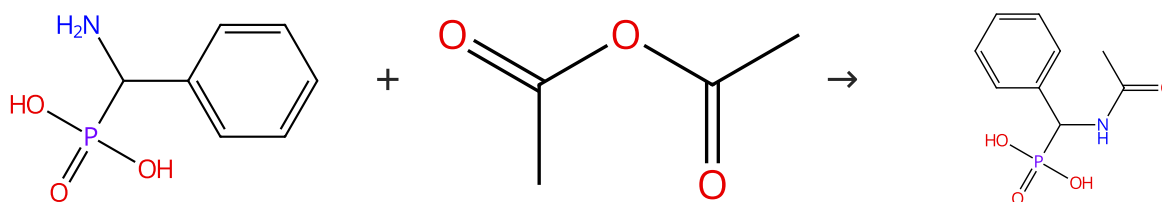
1.1 Solvents: Water; 30 min, 100 °C

By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (23)

Suppliers (84)

Suppliers (3)

31-366-CAS-4862818

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

1.1 Reagents: Acetic acid

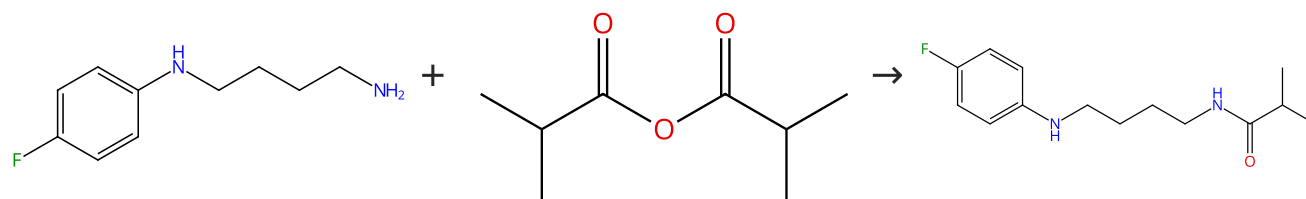
Solvents: Acetic acid; 30 min, 100 °C

By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

Suppliers (64)

Supplier (1)

31-366-CAS-8761444

Steps: 1 Yield: 100%

1,n-Diamines. Part 2: Synthesis of acyclic and hetero cyclic N-arylputrescine derivatives

1.1 Reagents: Sodium carbonate, Water
Solvents: Chloroform; 0 °C; 45 min, 0 °C

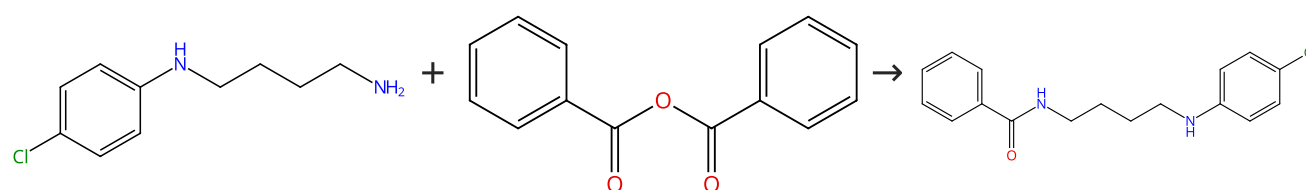
By: Diaz, Jimena E.; et al

Experimental Protocols

Tetrahedron Letters (2011), 52(16), 1895-1897.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

Suppliers (78)

Supplier (1)

31-366-CAS-4208193

Steps: 1 Yield: 100%

1,n-Diamines. Part 2: Synthesis of acyclic and hetero cyclic N-arylputrescine derivatives

1.1 Reagents: Sodium carbonate, Water
Solvents: Chloroform; 0 °C; 45 min, 0 °C

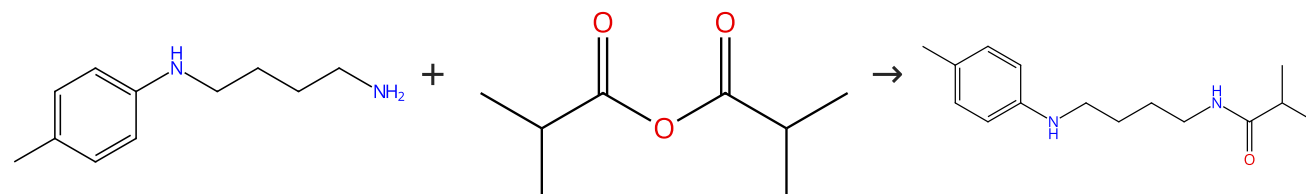
By: Diaz, Jimena E.; et al

Experimental Protocols

Tetrahedron Letters (2011), 52(16), 1895-1897.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

Suppliers (64)

Supplier (1)

31-366-CAS-10623473

Steps: 1 Yield: 100%

1,n-Diamines. Part 2: Synthesis of acyclic and hetero cyclic N-arylputrescine derivatives

1.1 Reagents: Sodium carbonate, Water
Solvents: Chloroform; 0 °C; 45 min, 0 °C

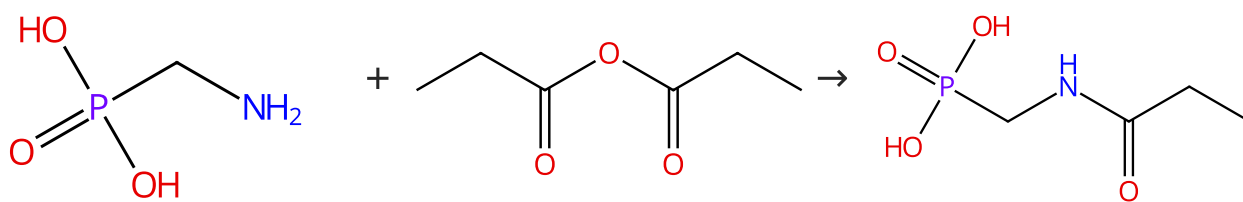
By: Diaz, Jimena E.; et al

Experimental Protocols

Tetrahedron Letters (2011), 52(16), 1895-1897.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (85)

Suppliers (47)

Supplier (1)

31-366-CAS-13377717

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

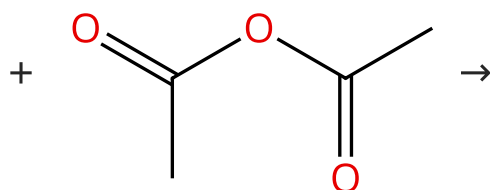
By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

1.1 Solvents: Water; 30 min, 100 °C

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
available in CAS
SciFinder

Suppliers (84)

Multi-component
structure image
available in CAS
SciFinder

31-366-CAS-1825854

Steps: 1 Yield: 100%

Sequential O- and N-Acylation Protocol for High-Yield Preparation and Modification of Rotaxanes: Synthesis, Functionalization, Structure, and Intercomponent Interaction of Rotaxanes

By: Tachibana, Yuya; et al

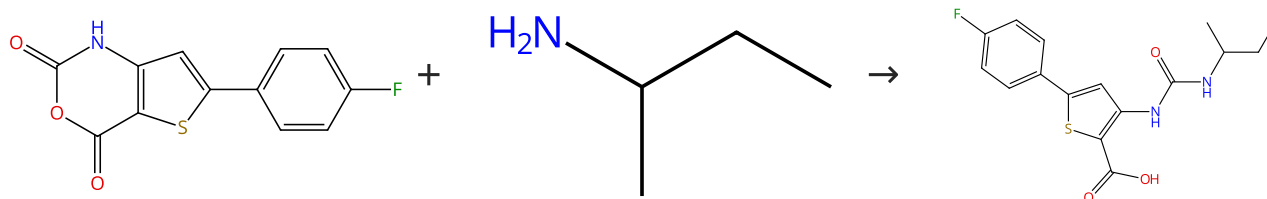
Journal of Organic Chemistry (2006), 71(14), 5093-5104.

1.1 Reagents: Triethylamine
Solvents: Acetonitrile; 192 h, 40 °C

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (71)

31-366-CAS-4088294

Steps: 1 Yield: 100%

Solution-phase parallel synthesis of a 1140-member ureidothiophene carboxylic acid library

By: Le Foulon, Francois-Xavier; et al

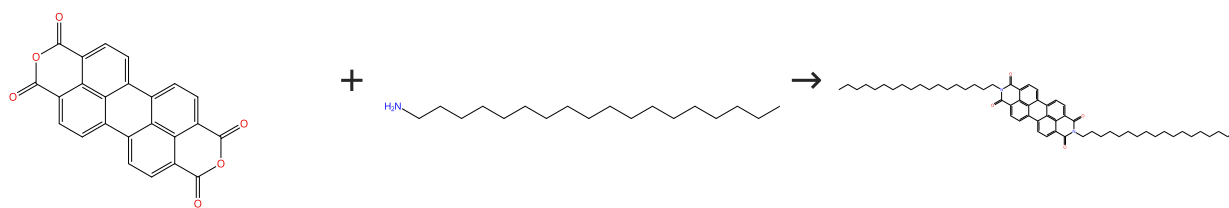
Journal of Combinatorial Chemistry (2005), 7(2), 253-257.

1.1 Solvents: Water; 2 h
1.2 Reagents: Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (85)

Suppliers (4)

31-366-CAS-16648088

Steps: 1 Yield: 100%

Green and highly efficient synthesis of perylene and naphthalene bisimides in nothing but water

1.1 Catalysts: Diisopropylethylamine
Solvents: Water; 20 min, rt; 4 h, 200 °C

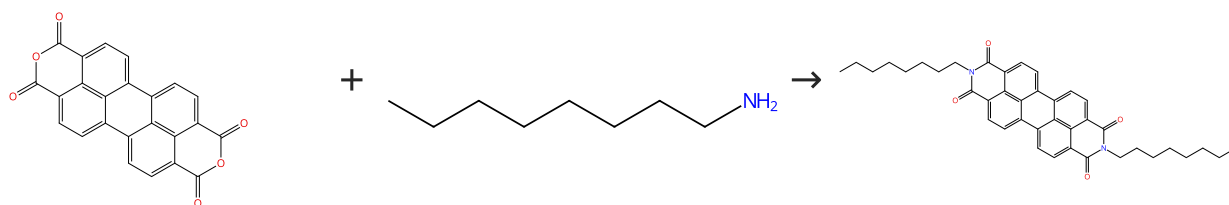
By: Baumgartner, Bettina; et al

Experimental Protocols

Chemical Communications (Cambridge, United Kingdom) (2017), 53(7), 1229-1232.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (79)

Suppliers (59)

31-366-CAS-16648080

Steps: 1 Yield: 100%

Green and highly efficient synthesis of perylene and naphthalene bisimides in nothing but water

1.1 Catalysts: Diisopropylethylamine
Solvents: Water; 20 min, rt; 4 h, 200 °C

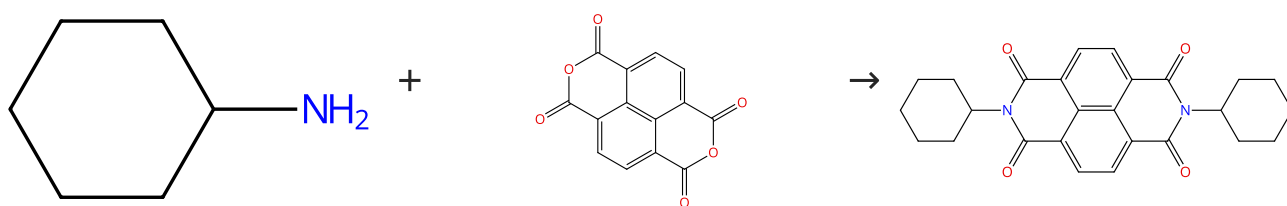
By: Baumgartner, Bettina; et al

Experimental Protocols

Chemical Communications (Cambridge, United Kingdom) (2017), 53(7), 1229-1232.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (85)

Suppliers (67)

Suppliers (17)

31-366-CAS-16648091

Steps: 1 Yield: 100%

Green and highly efficient synthesis of perylene and naphthalene bisimides in nothing but water

1.1 Catalysts: Diisopropylethylamine
Solvents: Water; 20 min, rt; 4 h, 200 °C

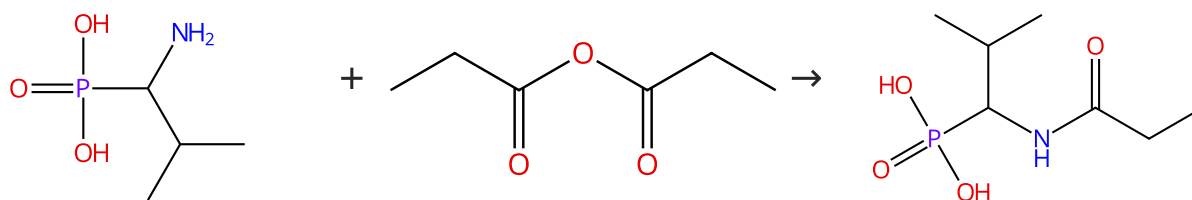
By: Baumgartner, Bettina; et al

Experimental Protocols

Chemical Communications (Cambridge, United Kingdom) (2017), 53(7), 1229-1232.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (24)

Suppliers (47)

Suppliers (3)

31-366-CAS-9951448

Steps: 1 Yield: 100%

1-(N-acylamino)alkanephosphonates. Part IV. N-acylation of 1-aminoalkanephosphonic acids

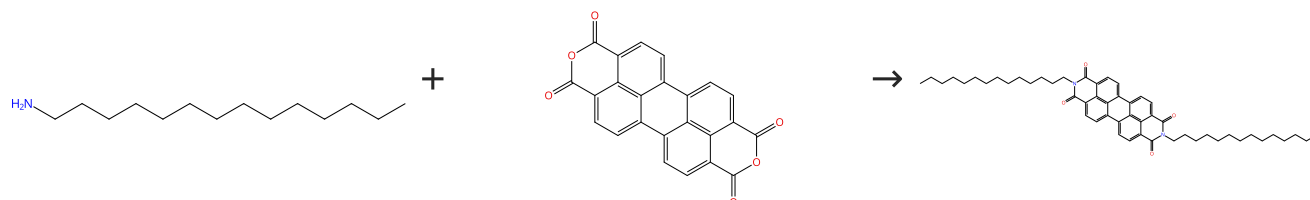
By: Kudzin, Z. H.; et al

Polish Journal of Chemistry (2005), 79(3), 499-513.

1.1 Solvents: Water; 30 min, 100 °C

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (64)

Suppliers (83)

31-366-CAS-16648085

Steps: 1 Yield: 100%

Green and highly efficient synthesis of perylene and naphthalene bisimides in nothing but water

By: Baumgartner, Bettina; et al

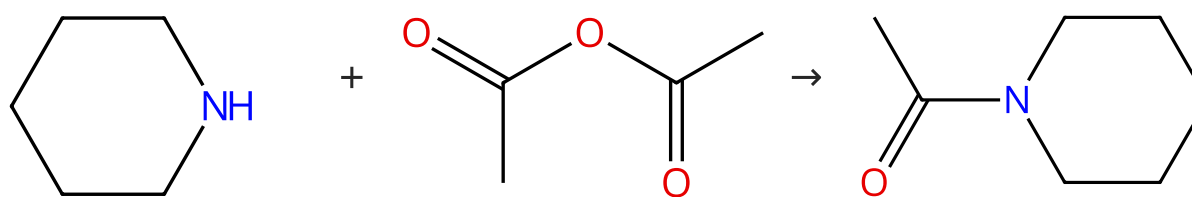
Chemical Communications (Cambridge, United Kingdom) (2017), 53(7), 1229-1232.

1.1 Catalysts: Diisopropylethylamine
Solvents: Water; 20 min, rt; 4 h, 200 °C

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (54)

Suppliers (84)

Suppliers (57)

31-366-CAS-11409203

Steps: 1 Yield: 100%

Highly efficient chemoselective N-acylation with water microreaction system in the absence of catalyst

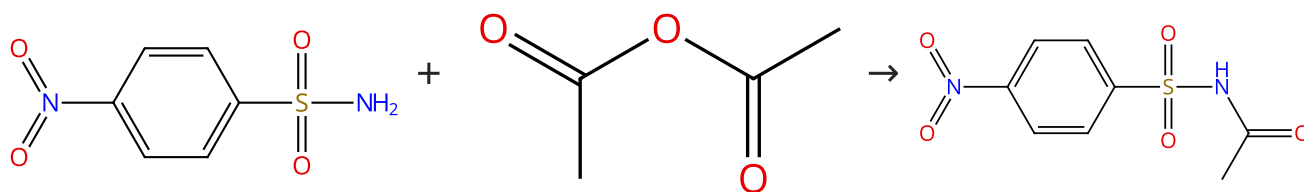
By: Sato, Masahiro; et al

Lab on a Chip (2009), 9(20), 2877-2880.

1.1 Reagents: Water; 1.1 s, 0.1 M Pa, 11 °C

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

Suppliers (84)

Suppliers (4)

31-366-CAS-9929941

Steps: 1 Yield: 100%

Green acylation of aromatic sulfonamides in heterogeneous catalysis

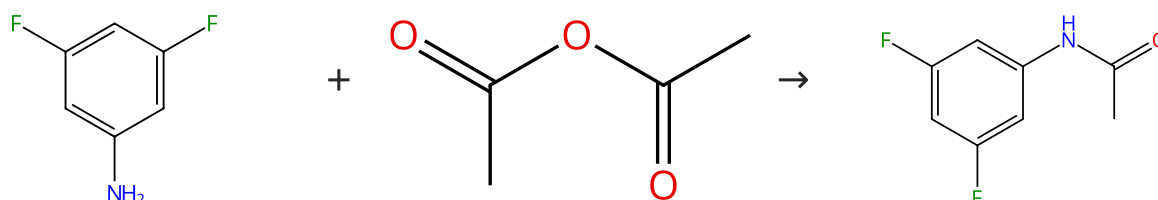
1.1 **Catalysts:** Fluorosulfuric acid
Solvents: Acetonitrile; 18 h, 80 °C

By: Coman, Simona M.; et al

Chemical Industries (Boca Raton, FL, United States) (2009), 123, 425-434.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (84)

Suppliers (52)

31-366-CAS-7944540

Steps: 1 Yield: 100%

Synthesis of Highly Functionalized 4-Aminoquinolines

1.1 **Reagents:** Triethylamine
Solvents: Dichloromethane; 4.5 h, rt; 17 h, rt

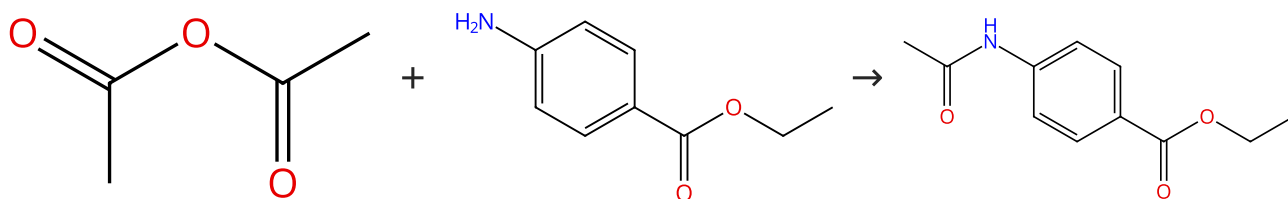
By: Wezeman, Tim; et al

Angewandte Chemie, International Edition (2016), 55(11), 3823-3827.

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (127)

Suppliers (66)

31-366-CAS-13011922

Steps: 1 Yield: 100%

Synthesis of Highly Functionalized 4-Aminoquinolines

1.1 **Reagents:** Triethylamine
Solvents: Dichloromethane; rt; 4.5 h, rt; 17 h, rt

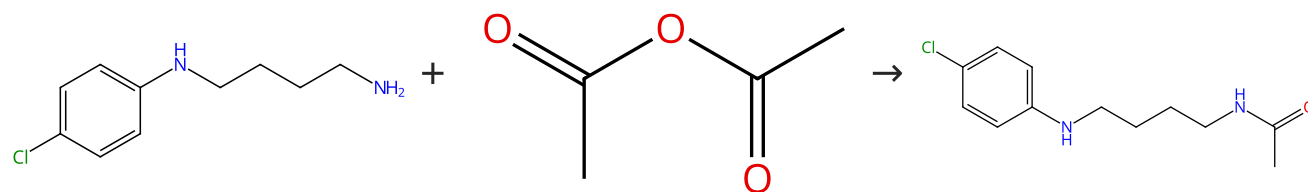
By: Wezeman, Tim; et al

Angewandte Chemie, International Edition (2016), 55(11), 3823-3827.

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

Suppliers (84)

Suppliers (2)

31-366-CAS-15471219

Steps: 1 Yield: 100%

1,n-Diamines. Part 2: Synthesis of acyclic and hetero cyclic N-arylputrescine derivatives

1.1 **Reagents:** Sodium carbonate, Water
Solvents: Chloroform; 0 °C; 45 min, 0 °C

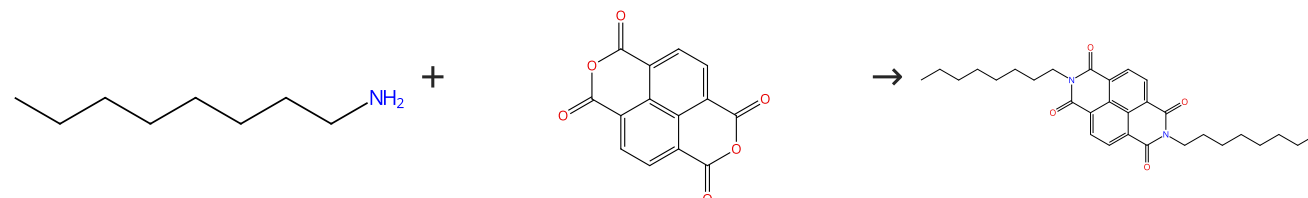
By: Diaz, Jimena E.; et al

Experimental Protocols

Tetrahedron Letters (2011), 52(16), 1895-1897.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (79)

Suppliers (67)

Suppliers (10)

31-366-CAS-16648089

Steps: 1 Yield: 100%

Green and highly efficient synthesis of perylene and naphthalene bisimides in nothing but water

1.1 **Catalysts:** Diisopropylethylamine
Solvents: Water; 20 min, rt; 4 h, 200 °C

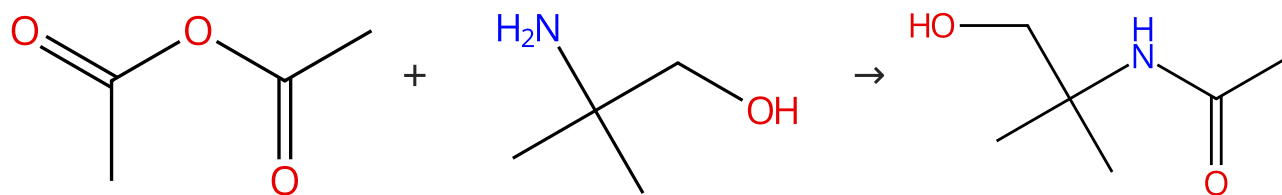
By: Baumgartner, Bettina; et al

Experimental Protocols

Chemical Communications (Cambridge, United Kingdom) (2017), 53(7), 1229-1232.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (93)

Suppliers (30)

31-366-CAS-3968091

Steps: 1 Yield: 100%

Nucleophilic acyl substitutions of anhydrides with protic nucleophiles catalyzed by amphoteric, oxomolybdenum species

1.1 **Catalysts:** Molybdenum chloride oxide (MoCl₂O₂), (7-4)-
Solvents: Dichloromethane; rt; 30 min, rt

By: Chen, Chien-Tien; et al

1.2 0.5 h, rt

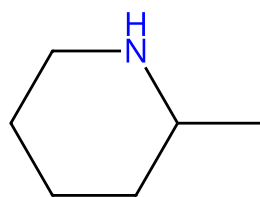
1.3 **Reagents:** Water; cooled

Journal of Organic Chemistry (2005), 70(4), 1188-1197.

Experimental Protocols

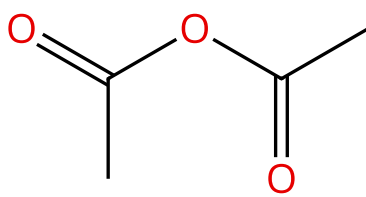
Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



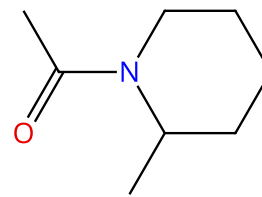
Suppliers (51)

+



Suppliers (84)

→



Suppliers (6)

31-366-CAS-13631026

Steps: 1 Yield: 100%

Highly efficient chemoselective N-acylation with water microreaction system in the absence of catalyst

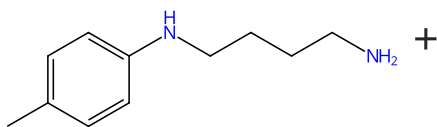
1.1 Reagents: Water; 1.1 s, 0.1 M Pa, 24 °C

By: Sato, Masahiro; et al

Lab on a Chip (2009), 9(20), 2877-2880.

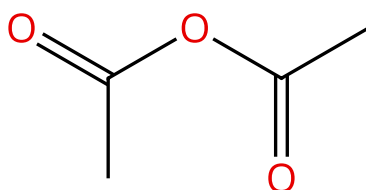
Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



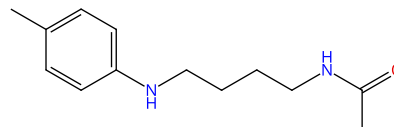
Suppliers (6)

+



Suppliers (84)

→



Supplier (1)

31-366-CAS-6346158

Steps: 1 Yield: 100%

1,n-Diamines. Part 2: Synthesis of acyclic and hetero cyclic N-arylputrescine derivatives
1.1 Reagents: Sodium carbonate, Water
Solvents: Chloroform; 0 °C; 45 min, 0 °C

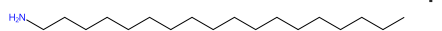
By: Diaz, Jimena E.; et al

Tetrahedron Letters (2011), 52(16), 1895-1897.

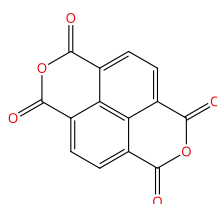
Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%

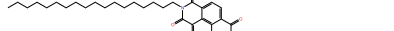


+



Suppliers (67)

→



Suppliers (6)

31-366-CAS-16648090

Steps: 1 Yield: 100%

Green and highly efficient synthesis of perylene and naphthalene bisimides in nothing but water
1.1 Catalysts: Diisopropylethylamine
Solvents: Water; 20 min, rt; 4 h, 200 °C

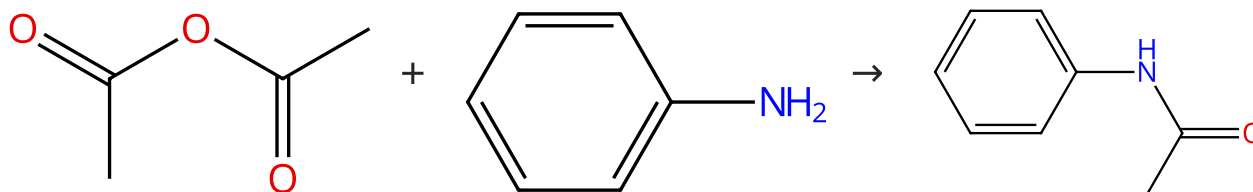
By: Baumgartner, Bettina; et al

Chemical Communications (Cambridge, United Kingdom) (2017), 53(7), 1229-1232.

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (113)

Suppliers (99)

31-366-CAS-16500482

Steps: 1 Yield: 100%

ZSM-5-SO₃H: An efficient catalyst for acylation of sulfonamides, amines, alcohols and phenols under solvent-free conditions

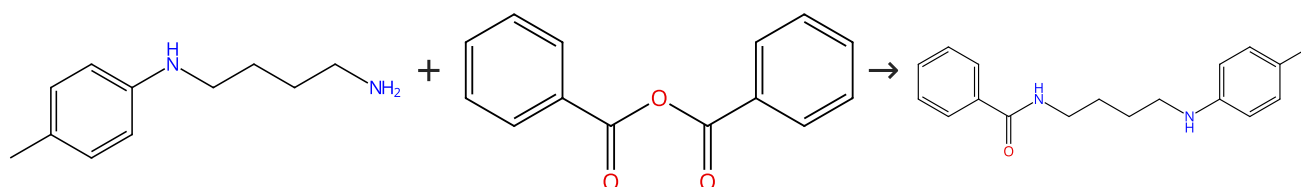
By: Massah, Ahmad Reza; et al

ISRN Organic Chemistry (2013), 951749/1-951749/12, 13 pp..

1.1 Reagents: Benzyl alcohol; 2 min, rt

Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

Suppliers (78)

Supplier (1)

31-366-CAS-12747643

Steps: 1 Yield: 100%

1,n-Diamines. Part 2: Synthesis of acyclic and hetero cyclic N-arylputrescine derivatives

By: Diaz, Jimena E.; et al

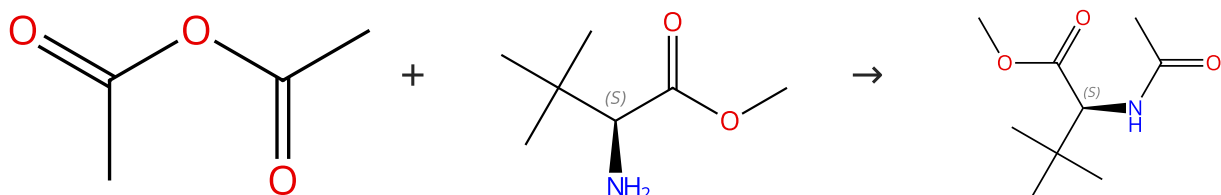
Tetrahedron Letters (2011), 52(16), 1895-1897.

1.1 Reagents: Sodium carbonate, Water
Solvents: Chloroform; 0 °C; 45 min, 0 °C

Experimental Protocols

Scheme 48 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (84)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (61)

Absolute stereochemistry shown

Suppliers (4)

31-366-CAS-13166374

Steps: 1 Yield: 100%

Nucleophilic acyl substitutions of anhydrides with protic nucleophiles catalyzed by amphoteric, oxomolybdenum species

By: Chen, Chien-Tien; et al

Journal of Organic Chemistry (2005), 70(4), 1188-1197.

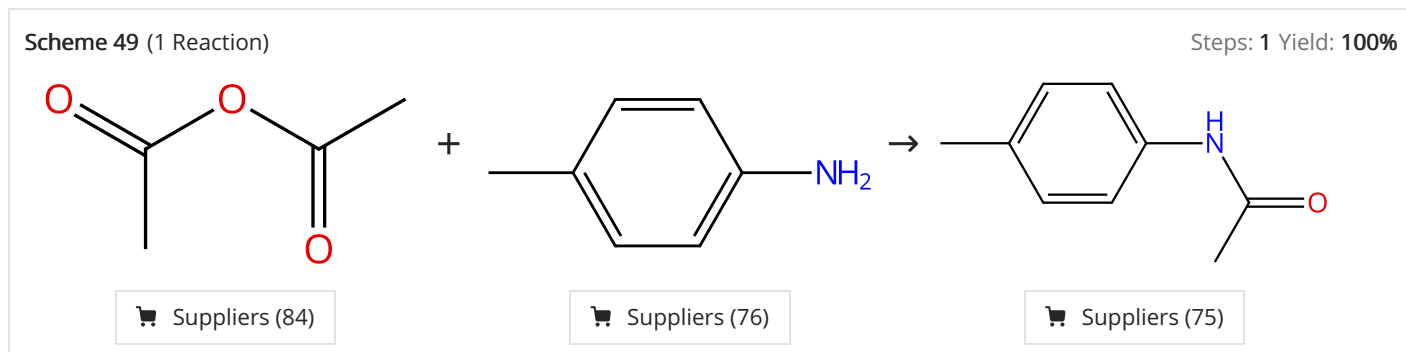
1.1 Catalysts: Molybdenum chloride oxide (MoCl₂O₂), (7-4)-
Solvents: Dichloromethane; rt; 30 min, rt

1.2 2 h, rt

1.3 Reagents: Water; cooled

Experimental Protocols

31-366-CAS-12355452 Steps: 1 Yield: 100% 1.1 Catalysts: Oxobis(1,1,1-trifluoromethanesulfonato-κO) vanadium Solvents: Dichloromethane 1.2 Reagents: Sodium bicarbonate Solvents: Water Experimental Protocols	Catalytic Nucleophilic Acyl Substitution of Anhydrides by Amphoteric Vanadyl Triflate By: Chen, Chien-Tien; et al Organic Letters (2001), 3(23), 3729-3732.
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31-366-CAS-16500485 Steps: 1 Yield: 100% 1.1 Reagents: Phenol; cooled; 2 min, rt	ZSM-5-SO₃H: An efficient catalyst for acylation of sulfonamides, amines, alcohols and phenols under solvent-free conditions By: Massah, Ahmad Reza; et al ISRN Organic Chemistry (2013), 951749/1-951749/12, 13 pp..
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